

LAOHEKOU, China

WMO: 572650

Lat: 32.4319N

Lon: 111.7319E

Elev: 129

StdP: 99.79

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.3	-2.2	-13.5	1.2	4.4	-10.5	1.6	4.1	6.9	4.8	5.6	7.1	1.2	70	0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	35.3	26.2	34.2	25.7	33.1	25.4	28.5	33.3	27.7	32.2	27.1	31.2	2.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.2	23.3	31.7	26.5	22.4	30.8	25.9	21.6	30.0	93.3	33.4	89.9	32.2	86.6	31.2	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	6.2	5.1	4.3	DB	-5.7	37.7	1.5	1.0	-6.7	38.4	-7.6	39.0	-8.4	39.5	-9.5	40.2
(5)				WB	-6.7	29.6	1.3	0.7	-7.6	30.1	-8.4	30.5	-9.1	30.9	-10.0	31.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.7	3.3	6.1	11.3	17.3	22.3	26.1	28.3	27.4	23.0	17.8	11.2	5.6
	DBStd	9.16	2.87	3.60	3.98	3.81	3.11	2.71	2.41	2.89	3.17	3.28	3.69	2.79
	HDD10.0	523	207	116	32	2	0	0	0	0	0	28	139	
	HDD18.3	1756	465	343	219	62	5	0	0	3	49	214	396	
	CDD10.0	2966	1	7	72	221	381	483	567	538	389	242	64	2
	CDD18.3	1158	0	0	2	31	128	233	308	280	143	33	1	0
	CDH23.3	11139	0	0	19	248	1011	2264	3459	2943	998	189	8	0
	CDH26.7	4543	0	0	1	62	369	952	1523	1257	342	37	0	0
Wind	WSAvg	1.9	1.7	1.9	2.2	2.1	2.1	2.0	1.9	1.8	1.8	1.7	1.7	1.7
Precipitation	PrecAvg	846	20	29	52	80	98	82	114	122	115	72	46	17
	PrecMax	1205	82	109	114	261	235	226	338	302	321	259	142	60
	PrecMin	474	0	0	4	13	35	11	24	24	10	4	3	0
	PrecStd	195	17	24	27	51	47	52	73	72	81	51	35	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.2	19.1	25.8	31.3	34.7	36.1	37.0	36.6	34.7	30.2	24.4	17.3
		MCWB	7.6	10.9	15.6	19.8	21.0	23.7	28.0	27.0	24.2	19.6	15.3	9.0
	2%	DB	12.4	16.4	22.6	28.3	32.3	34.5	35.4	34.9	32.1	27.4	21.6	14.6
		MCWB	6.5	9.8	14.7	19.3	21.2	23.9	27.7	26.9	23.4	18.3	14.2	8.3
	5%	DB	10.3	14.3	20.3	26.2	30.6	33.2	34.1	33.6	30.3	25.4	19.1	12.5
		MCWB	5.5	8.5	13.7	18.2	20.8	24.0	27.2	26.3	22.2	17.5	13.0	7.2
	10%	DB	8.4	12.2	18.1	24.1	28.7	31.6	32.9	32.3	28.5	23.4	17.0	10.6
		MCWB	4.6	7.4	12.3	17.3	20.3	23.6	26.8	25.9	21.5	17.1	12.4	6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	12.6	18.4	22.9	24.7	27.6	29.6	29.2	27.1	21.4	16.6	10.7
		MCDB	13.6	16.9	22.6	27.9	29.8	32.9	34.5	34.4	32.3	25.6	22.0	13.8
	2%	WB	7.2	10.6	16.1	20.8	23.2	26.7	28.8	28.2	25.3	20.1	15.5	9.2
		MCDB	11.1	15.1	20.8	25.9	28.7	31.6	33.8	32.6	29.6	24.9	19.6	12.9
	5%	WB	6.1	9.3	14.4	19.5	22.2	25.8	28.1	27.5	23.9	19.1	14.2	8.1
		MCDB	9.4	12.9	18.9	24.5	27.5	30.2	32.8	31.7	27.9	23.2	17.6	11.5
	10%	WB	4.9	8.2	13.0	18.1	21.4	24.9	27.4	26.9	22.7	18.2	13.1	7.0
		MCDB	7.9	11.6	17.5	22.7	26.5	29.1	31.7	30.8	26.6	21.8	16.3	10.0
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.3	8.9	9.6	9.6	8.5	7.2	7.3	7.6	8.0	8.0	7.8
		MCDBR	11.9	12.7	13.2	13.6	13.4	11.3	9.3	9.5	10.9	12.0	12.3	12.4
		MCWBR	7.0	7.1	6.6	6.2	4.8	3.9	3.5	3.2	3.7	4.5	5.8	7.0
	5% WB	MCDBR	10.9	11.1	11.7	12.0	10.4	8.8	8.4	8.2	8.8	9.4	10.4	10.8
		MCWBR	6.9	6.8	6.5	6.5	4.8	3.9	3.6	3.4	4.0	4.3	5.5	6.6
Clear-Sky Solar Irradiance	taub	0.642	0.708	0.701	0.688	0.670	0.724	0.711	0.750	0.728	0.704	0.680	0.624	
	taud	1.537	1.454	1.492	1.555	1.635	1.572	1.630	1.539	1.550	1.531	1.537	1.586	
	Ebn at Noon	542	559	617	656	675	637	643	607	592	555	510	526	
	Edh at Noon	231	276	284	276	257	273	257	277	264	251	229	210	
All-Sky Solar Radiation	RadAvg	2.23	2.56	3.69	4.41	4.65	4.79	4.63	4.21	3.42	2.95	2.38	2.12	
	RadStd	0.34	0.58	0.38	0.44	0.47	0.47	0.64	0.60	0.46	0.47	0.39	0.37	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	-0.42	-1.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	+0.38	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page